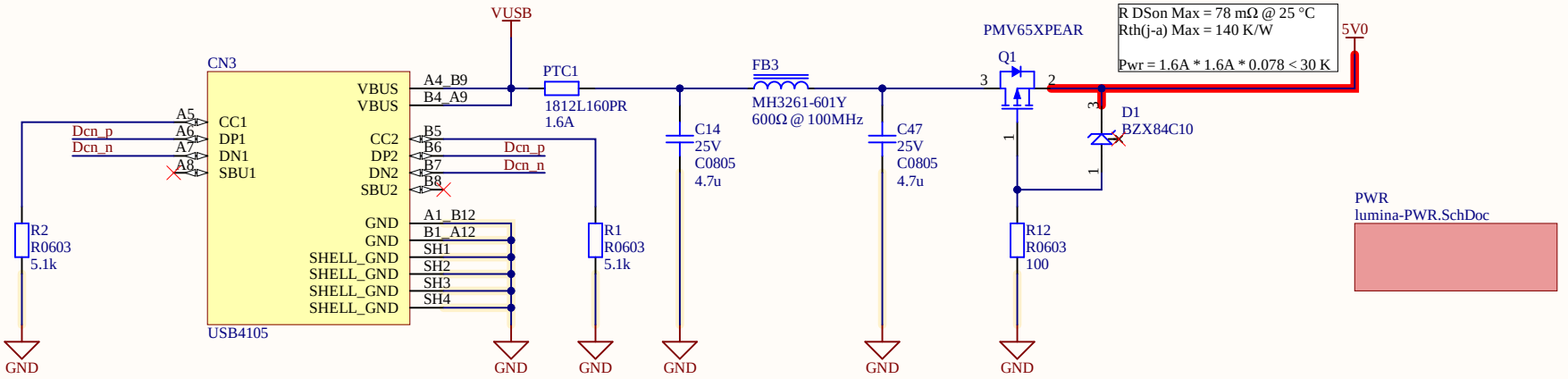
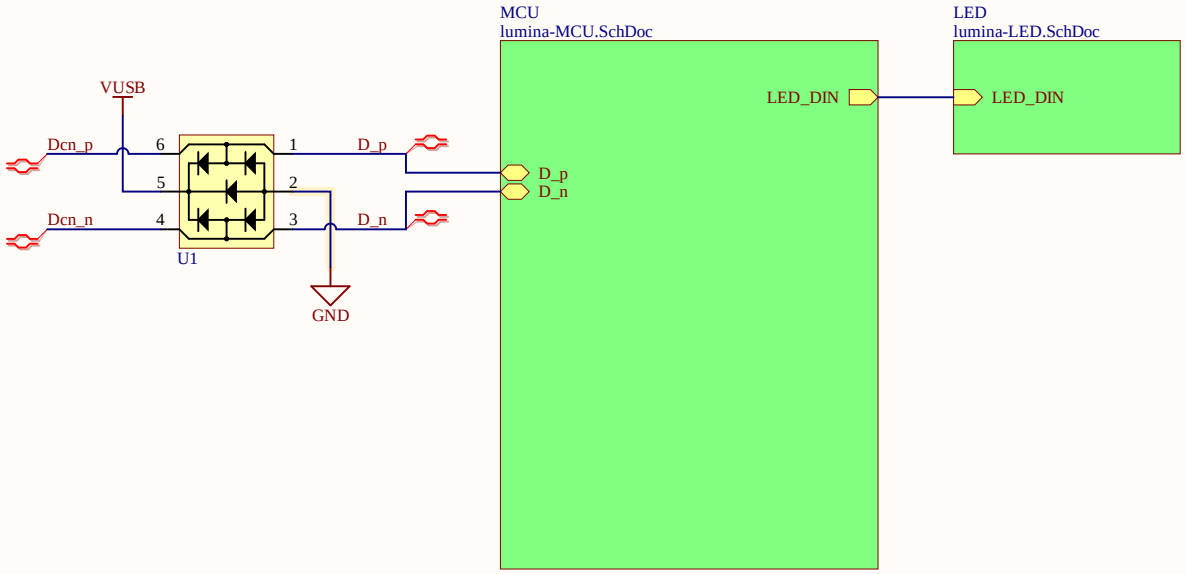
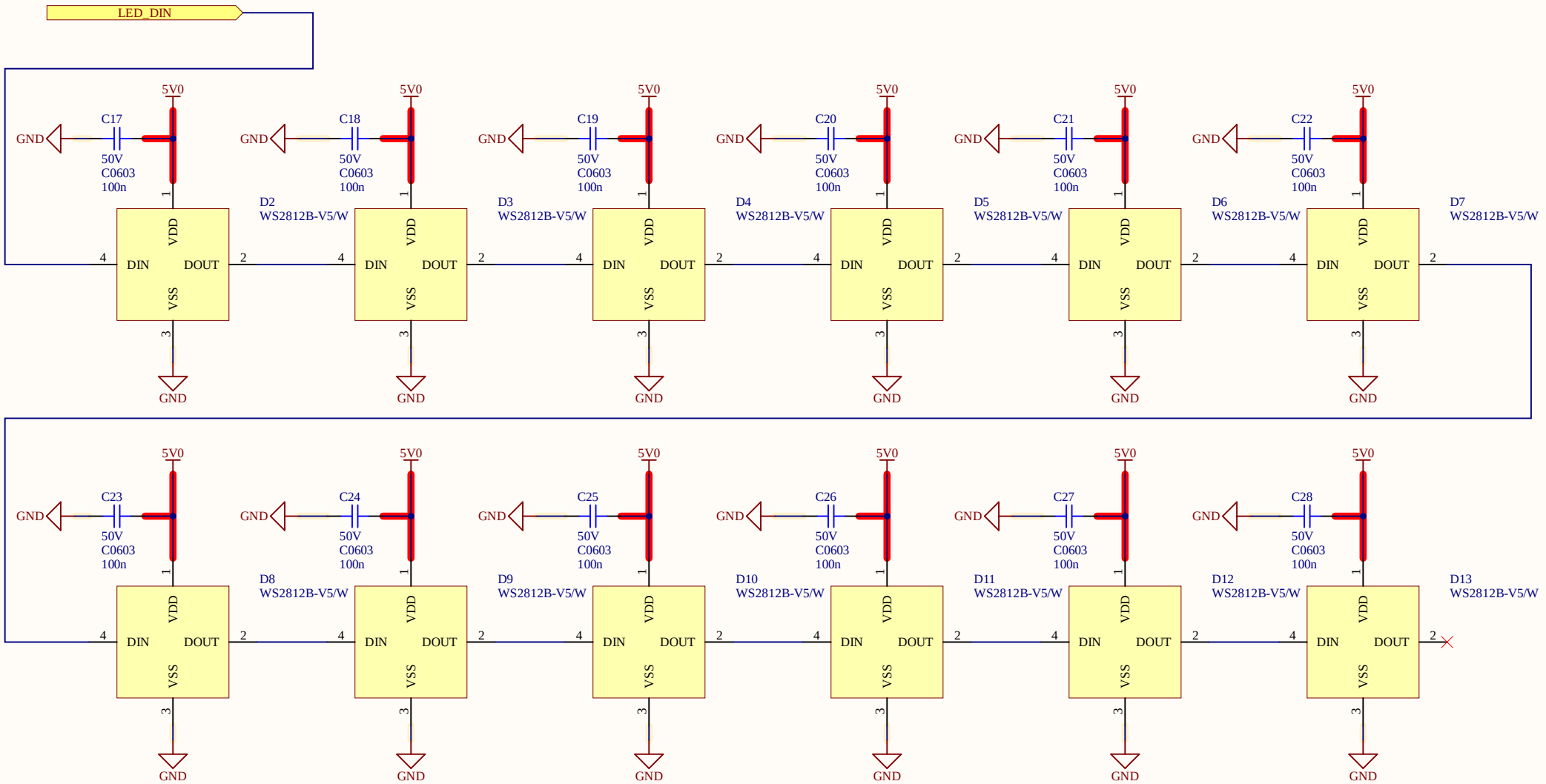
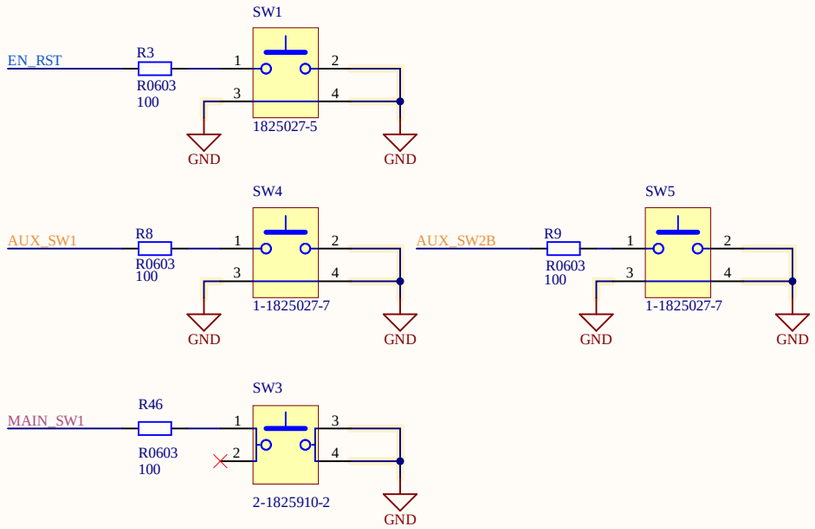
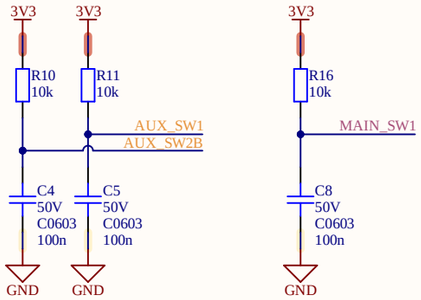

VUSB
CN3
VBUS
VBUS
CC1
DP1
DN1
SBU1
CC2
DP2
DN2
SBU2
GND
GND
SHELL_GND
SHELL_GND
SHELL_GND
SHELL_GND
USB4105
A4_B9
B4_A9
B5
B6
B7
B8
A1_B12
B1_A12
SH1
SH2
SH3
SH4
A5
A6
A7
A8
Dcn_p
Dcn_n
Dcn_p
Dcn_n
PTC1
1812L160PR
1.6A
FB3
MH3261-601Y
600Ω @ 100MHz
PMV65XPEAR
Q1
R Dson Max = 78 mΩ @ 25 °C
Rth(j-a) Max = 140 K/W
Pwr = 1.6A * 1.6A * 0.078 < 30 K
5V0
D1
BZX84C10
R2
R0603
5.1k
R1
R0603
5.1k
C14
25V
C0805
4.7u
C47
25V
C0805
4.7u
R12
R0603
100
PWR
lumina-PWR.SchDoc
GND

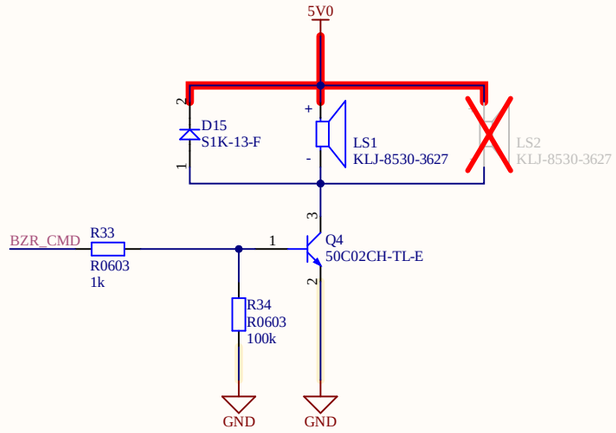

MCU
lumina-MCU.SchDoc
LED
lumina-LED.SchDoc
LED_DIN
LED_DIN
VUSB
Dcn_p
Dcn_n
6
5
4
1
2
3
D_p
D_n
U1
D_p
D_n
GND

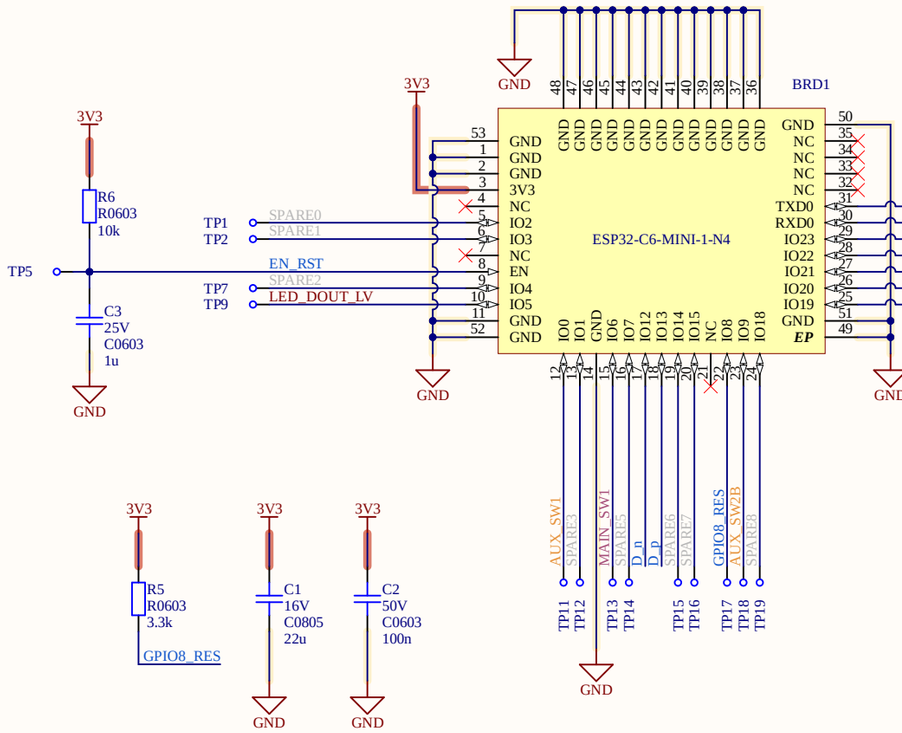
FID1 FID2

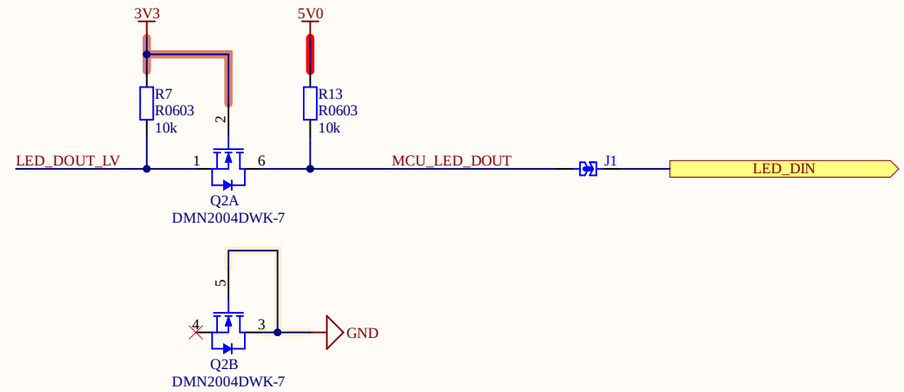

LED_DIN
5V0
GND
C17
C18
C19
C20
C21
C22
C23
C24
C25
C26
C27
C28
50V
C0603
100n
D2
WS2812B-V5/W
D3
D4
D5
D6
D7
D8
D9
D10
D11
D12
D13
VDD
DIN
DOUT
VSS

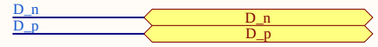

SW1
EN_RST
R3
R0603
100
1825027-5
GND
SW4
AUX_SW1
R8
R0603
100
1-1825027-7
SW5
AUX_SW2B
R9
R0603
100
1-1825027-7
SW3
MAIN_SW1
R46
R0603
100
2-1825910-2


3V3
R10
10k
R11
10k
AUX_SW1
AUX_SW2B
C4
50V
C0603
100n
C5
50V
C0603
100n
R16
10k
MAIN_SW1
C8
50V
C0603
100n


5V0
D15
S1K-13-F
LS1
KLJ-8530-3627
LS2
KLJ-8530-3627
BZR_CMD
R33
R0603
1k
Q4
50C02CH-TL-E
R34
R0603
100k
GND


BRD1
ESP32-C6-MINI-1-N4
3V3
GND
R6
R0603
10k
C3
25V
C0603
1u
TP1
TP2
TP5
TP7
TP9
SPARE0
SPARE1
EN_RST
SPARE2
LED_DOUT_LV
SPARE14
SPARE13
BZR_CMD
SPARE12
SPARE11
SPARE10
SPARE9
TP20
TP21
TP3
TP4
TP6
TP8
TP10
AUX_SW1
SPARE3
MAIN_SW1
SPARE5
D_n
D_p
SPARE6
SPARE7
GPIO8_RES
AUX_SW2B
SPARE8
TP11
TP12
TP13
TP14
TP15
TP16
TP17
TP18
TP19
R5
R0603
3.3k
GPIO8_RES
C1
16V
C0805
22u
C2
50V
C0603
100n


3V3
5V0
R7
R0603
10k
R13
R0603
10k
LED_DOUT_LV
MCU_LED_DOUT
J1
LED_DIN
Q2A
DMN2004DWK-7
Q2B
DMN2004DWK-7
GND


D_n
D_p
D_n
D_p

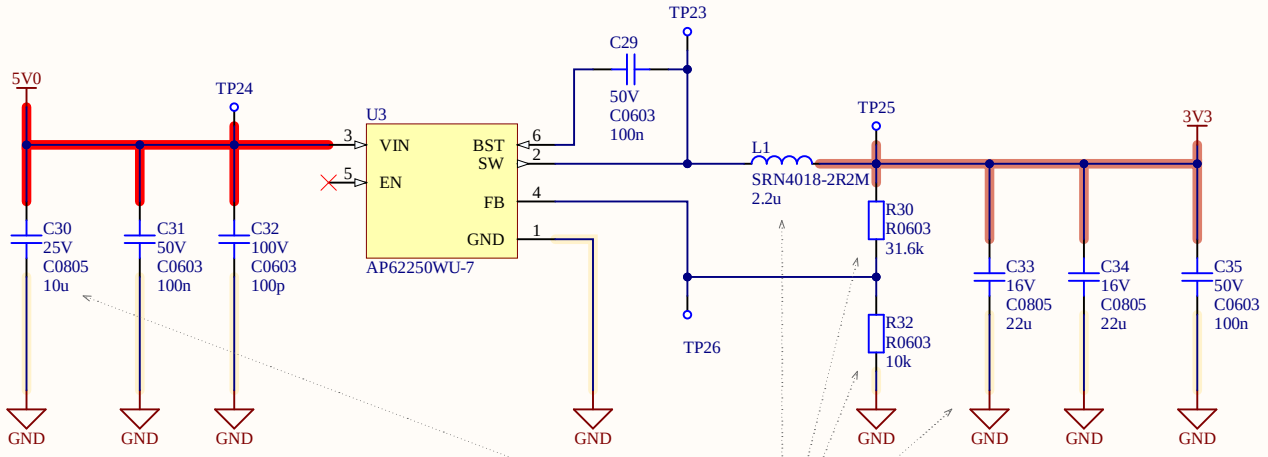


Table 1. Recommended Component Selections

AP62250						
Output Voltage (V)	R1 (kΩ)	R2 (kΩ)	L (µH)	C1 (µF)	C2 (µF)	C3 (nF)
1.2	4.99	10	1.0	10	2 x 22	100
1.5	8.66	10	1.0	10	2 x 22	100
1.8	12.4	10	1.5	10	2 x 22	100
2.5	21.5	10	1.5	10	2 x 22	100
3.3	31.6	10	2.2	10	2 x 22	100
5.0	52.3	10	2.2	10	2 x 22	100